A Deterministic Brane-Lattice Substrate for Emergent Spacetime, Gauge Structure, and Matter

Geometric Nonlinearity in a Flat 4D Euclidean Ambient, and a Retrocausal Worldtube Reading of Bell

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Abstract

The Standard Model and general relativity are extraordinarily successful effective descriptions of observed physics, yet they leave open a complementary explanatory question: whether a smaller set of underlying ingredients could give rise to relativistic kinematics, gauge structure, and particle-like localization as emergent regularities. This paper develops that question in a deterministic *substrate-first* setting.

We study a brane lattice with the ontology of a 4D cubic lattice embedded in a 4D Euclidean ambient with codimension zero; the ambient is fully symmetric, and the asymmetry that picks one of the four lattice directions as time lives in the brane action's Lorentzian sign structure rather than in the ambient geometry. For technical work the brane is sliced perpendicular to its timelike direction, giving an equivalent spacelike-slice description with material domain $\Omega \subset \mathbb{R}^3$ and slicing parameter t, on which the long-wavelength effective theory is an embedding field $\mathbf{X} : \Omega \times \mathbb{R} \to \mathbb{R}^4$ governed by an isotropic hyperelastic action built from the induced metric. A homogeneous mismatch parameter α encodes the prestretch between the held lattice spacing and the stress-free link length. A central modeling point is that the essential nonlinearity is *geometric*, not imposed by an auxiliary field or by hand-written thresholds: even when the microscopic link energy is Hooke-linear in the scalar extension, the 4D Euclidean link length and the induced metric make the map from substrate positions to forces nonlinear. This same geometric mechanism couples transverse amplitude to in-brane contraction.

In the weak, narrowband regime, linearization yields polarized branches with characteristic wave cones. When one branch dominates and dispersion is negligible, observers built from that same branch inherit an effective Lorentz-type kinematics. A slowly varying transverse excitation also modifies intrinsic distances and, under prestretch, produces an inward lateral response; in a weak-field regime we interpret this as a gravity-like contraction channel. For each band-isolated wavepacket, adiabatic transport of its locally retained polarization subspace induces Berry (rank-1) or Wilczek–Zee (non-Abelian) connections, which we interpret as the natural gauge variables of the per-wavepacket coarse-grained envelope theory. On a cubic lattice, an axis-aligned triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$ provides a natural candidate internal sector: in the decoupled limit it supports three approximately Abelian phases, while finite inter-axis mixing promotes the description to a non-Abelian $U(3)$ transport problem with an $SU(3)$-like traceless part.

Localized particle-like states are posed as confined wave modes rather than point objects. Linearization about symmetric backgrounds yields self-adjoint eigenproblems with spherical-harmonic angular structure, while nonlinear self-guidance is attributed to the same induced-metric geometry that couples amplitude to lateral strain. This paper therefore has two aims: first, to consolidate the core theory into a coherent lattice-to-continuum statement; second, to define the decisive numerical program. Because the cubic substrate naturally singles out a three-channel internal sector, the next simulation priority is argued to be a baryon-like triplet mode (proton/neutron sector) rather than another electron-only ansatz. The intended contribution is not a finished replacement for established theories, but a sharpened substrate model whose mathematical bridges are explicit enough to be attacked directly in simulation.

We close by committing to an interpretive stance dictated by Bell's theorem: causality is a property of soliton chirality rather than of the substrate, and entanglement is the continuity of one branching 4D worldtube — a retrocausal completion in the same family as the two-state-vector, transactional, and time-symmetric Bohmian readings.

1 Introduction

The Standard Model and general relativity are extraordinarily successful at organizing experimental facts. At the same time, their combined fundamental inventory—multiple fields, symmetry sectors, symmetry breaking structure, and many parameters fixed by measurement—invites a complementary line of inquiry: whether a smaller set of ground ingredients could underlie this effective complexity. This paper explores such a *minimal-ingredient reconstruction* program.

Substrate-first strategy. We adopt a deterministic *substrate-first* strategy: posit an ontically real wave medium and treat familiar field theories as coarse-grained descriptions of constrained wave dynamics. The goal is *not* to dispute established phenomenology, but to ask what could make gauge structure, relativistic kinematics, and particle-like quantization appear naturally as emergent regularities.

Ground ingredients (minimal commitments). The theoretical framework developed here keeps its primitive commitments deliberately small: (i) a single substrate ontologically realised as a 4D cubic lattice embedded with codimension zero in a 4D Euclidean ambient, with the timelike direction picked out by the Lorentzian sign structure of the brane action (Section 3.1), and macroscopically described by its continuum long-wavelength theory on a spacelike slice, (ii) pure geometric coupling between displacements perpendicular to a link and the lengthening of that link, and (iii) a controlled separation between a linear wave sector and a nonlinear self-guidance sector. Gauge potentials, relativistic coordinates, and particle labels are then interpreted as emergent bookkeeping attached to stable excitation sectors rather than as independent fundamental substances.

Geometric source of nonlinearity. One modeling distinction is crucial throughout this paper. The microscopic link law may be linear in the scalar extension, but the extension itself is the full Euclidean distance between neighboring nodes in $\mathbb{R}^4$. Equivalently, in the continuum description the induced metric already contains the quadratic term $\partial_i u\, \partial_j u$ when the fourth embedding component is excited. The primary nonlinearity of the theory therefore comes from geometry—the Pythagorean dependence of link lengths and the induced metric—not from an added interaction field, an imposed threshold, or a hand-crafted saturation rule. This point matters because the same geometric mechanism is intended to underwrite contraction, self-guidance, and localized particle structure.

Substrate hypothesis. We assume a brane lattice with the 4D-in-4D ontology of Section 3.1: a 4D cubic lattice in a symmetric 4D Euclidean ambient, with the timelike direction selected by the brane action's Lorentzian sign structure. For technical analysis on a spacelike slice the brane is parameterised by an embedding map

$$\mathbf{X} : \Omega \times \mathbb{R} \to \mathbb{R}^4, \qquad (x, t) \mapsto \mathbf{X}(x, t), \qquad \Omega \subset \mathbb{R}^3, \tag{1}$$

with t the slicing parameter along the inside observer's timelike direction. The dynamics is specified by a local hyperelastic action on Ω, constructed from invariants of the induced metric of the embedding. This is the minimal coordinate-free way to write an elastic energy without hard-coding preferred directions into the continuum description; microscopically, a cubic lattice substrate is allowed, and large-scale isotropy is treated as an empirical constraint on admissible lattice couplings and parameter regimes [17, 30, 38].

Gravity from transverse excitation (contraction). A central motivating idea is that excitation in the embedding dimension generically feeds back into *lateral* stresses. If a region of the brane develops transverse structure $X^4 = u(x,t)$, then the induced metric gains the quadratic contribution $\partial_i u\, \partial_j u$. Because the in-brane displacement is itself dynamical, a prestretched medium relaxes by generating an inward-biased lateral response rather than only a local stress increase. Coarse-grained over scales large compared to the carrier wavelength, such slowly varying contraction fields can bias the propagation of other wave packets and the equilibrium of localized modes. We therefore treat Newtonian gravity as an emergent, long-range *contraction channel* mediated by transverse excitation, not as a separately postulated fundamental field.

Per-wavepacket band isolation and geometric phase. The complex-envelope description that promotes the real lateral triplet to $\Psi \in \mathbb{C}^3$ does not require a globally coherent carrier sector spanning macroscopic scales. Instead it is applied per band-isolated excitation: each wavepacket carries its own local carrier frequency ω_0 and its own retained polarization subspace, defined as an eigenbundle of the linearized carrier operator about the slowly varying background at the wavepacket's location. Adiabatic transport of that local subspace induces a Berry (rank-1) or Wilczek–Zee (non-Abelian) connection, which is interpreted as the natural origin of gauge variables in the per-wavepacket effective description [9, 10, 12, 39, 51].

Microscopic cubicity and macroscopic isotropy. The cubic lattice hypothesis is not rotationally symmetric on the microscopic level. The central question is whether long-wavelength observers built from the same substrate could nonetheless inherit a universal effective metric — and whether birefringence or residual direction dependence survives as a decisive diagnostic if they do not. The full dual-observer argument (lab anisotropy vs internal isotropy) is stated canonically in Section 3.2.

Particles as solitons; probability not fundamental. We do not take the probabilistic interpretation of quantum theory as fundamental. Instead, apparent point-like interactions and discrete outcomes are attributed to deterministic wave dynamics combined with Fourier-localization constraints and nonlinear exchange events. Particles are not fundamental point objects in this framework; they are localized standing-wave patterns of the substrate, potentially with internal polarization and topological structure. The spin-$\frac{1}{2}$ 4π aspect is attributed to geometric phase/holonomy of an internal polarization subspace rather than to literal multi-winding of a photon path.

Relation to existing ideas. The emergence of an effective light cone and Lorentz-like kinematics from a substrate whose time direction is selected by the action signature rather than by ambient geometry has close analogues in analogue-gravity systems [7, 33, 50]. The present use of the term "brane" is literal (an elastic medium), and should not be confused with field-theoretic brane-world scenarios in high-energy theory [42]. The closest contemporary programme is the quaternionic quantum mechanics of Danielewski, Sapa, and Roth, in which the vacuum is an ideal elastic solid (a Planck–Kleinert crystal), elementary particles are soliton-like standing waves, gravity appears as a density-dependent change in wave speed (an effective index of refraction), and baryons are modelled as three-quark standing-wave complexes [15, 16]. We share the elastic-medium ontology and the particles-as-solitons commitment, but differ in three structural choices: the carrier here is a *real* vector substrate on a 4D codimension-zero brane whose time direction is selected by the action signature rather than an external time; the complex and gauge structure is *derived* per wavepacket from band isolation and Berry/Wilczek–Zee holonomy rather than postulated through a quaternion field; and we make no claim to fix fundamental constants from the model.

Paper organization. Section 2 states the precise scope and the numerical priorities. Section 3 formulates the lattice and continuum substrate model, making the geometric source of the nonlinearity explicit. Section 4 discusses the linear-wave regime and an emergent effective light cone. Sections 5 and 6 develop the per-wavepacket band-isolation framing and the associated Berry/Wilczek–Zee gauge structure. Section 7 formulates localized particle modes as self-adjoint eigenproblems with nonlinear self-guidance. Section 8 states the interpretive commitment of the theory under Bell's theorem: a retrocausal worldtube reading in which causality emerges from soliton chirality and entanglement is the continuity of one branching 4D worldtube. Section 9 then identifies the most decisive next step: a baryon-like triplet simulation on the cubic substrate, where the axis-aligned internal structure is expected to matter most directly.

2 Scope, contributions, and non-claims

This paper develops a deterministic substrate theory with a narrow purpose: to state the core lattice-to-continuum theory coherently, to identify which bridges are already fixed by the model, and to isolate the decisive computations that must now be carried out. A cubic lattice is treated as the candidate microphysical substrate, while the continuum formulation is retained as the long-wavelength effective description and as the cleanest way to write isotropic constitutive baselines.

2.1 Positioning relative to established physics

The Standard Model and general relativity are treated here as empirically validated effective frameworks. Nothing in this paper is intended to weaken confidence in their experimental domain of applicability. The present work instead targets a different explanatory layer: it asks whether familiar structures such as effective Lorentz symmetry, gauge connections, and quantized particle spectra could arise as emergent regularities from a smaller set of ground ingredients.

2.2 Core theory delivered here

The theoretical framework developed here is organized around six coupled elements:

1. **A single substrate with two descriptions.** A 4D cubic lattice embedded with codimension zero in a 4D Euclidean ambient is taken as the microphysical substrate; the timelike direction is selected by the Lorentzian sign structure of the brane action (Section 3). For wavelengths large compared to the lattice spacing, its dynamics on a spacelike slice admits a continuum approximation on an intrinsic 3-domain $\Omega \subset \mathbb{R}^3$. An isotropic hyperelastic action remains the analytic baseline, but the current cubic stencil has a measurable leading-order anisotropy unless its shell weights are retuned.

2. **Geometric nonlinearity as the primary coupling mechanism.** The central amplitude–lateral coupling does not come from an auxiliary field. It comes from the 4D Euclidean link length on the lattice and, equivalently, from the induced metric in the continuum. This geometric mechanism is the intended source of transverse contraction, self-guidance, and localization.

3. **A prestretched substrate with an emergent gravity-like channel.** A homogeneous mismatch parameter α encodes the prestretch between held spacing and stress-free length. Because transverse gradients modify intrinsic distances, localized transverse excitation produces an inward lateral pull (contraction) and a slowly varying strain field. In a weak,

slowly varying regime this is interpreted as the qualitative origin of a long-range gravity-like channel (Section 3.9).

4. **Per-wavepacket band isolation and complex-envelope promotion.** The operational ingredient is not a globally coherent carrier sector but per-wavepacket band isolation: each excitation is treated as narrowband around its own local carrier frequency ω_0, with its own retained polarization subspace defined as an eigenbundle of the linearized carrier operator about the slowly varying background at the wavepacket's location. This local band isolation enables the controlled multiscale envelope description and promotes the real lateral triplet to $\Psi \in \mathbb{C}^3$ (Sections 5 and 6). No universe-wide coherent carrier is postulated.

5. **Berry/Wilczek–Zee transport as emergent gauge structure.** Adiabatic transport of a retained polarization subspace induces a Berry connection (rank-1) or a Wilczek–Zee connection (non-Abelian). These are interpreted as the natural gauge variables of the slow-sector description, rather than as independent microscopic fields (Sections 5 and 6).

6. **Localized particle modes and the baryon priority.** Particle-like states are modeled as localized standing-wave/solitonic modes whose spatial structure is organized by symmetry and whose amplitude scale is fixed by nonlinear self-guidance and energy normalization (Section 7). Because the cubic substrate naturally singles out an axis-aligned triplet sector, the first decisive non-electron target is argued to be a baryon-like confined triplet mode, with proton/neutron structure treated as the primary numerical priority.

2.3 Non-claims

To keep the scope honest, several items are explicitly *not* claimed in this manuscript:

- a derivation of the full Standard Model gauge group, generations, or precise coupling constants,
- a completed quantitative reduction of the contraction channel to a Newtonian limit with a derived value of G,
- a completed derivation of Maxwell or Yang–Mills dynamics in all regimes,
- a finished proton or neutron solution,
- a closed multi-particle theory including scattering cross sections, bound-state spectra, and entanglement phenomenology,
- an identification of any specific topological object (vortex ring, texture, etc.) with a specific physical particle: Section 6.3 records the topological structure of each sector as a mathematical fact, but which objects the substrate dynamics stabilizes is open (`OPEN_PROBLEMS.md` D4),
- a derivation of the QCD-vs-EM scale hierarchy from the prestress parameter α: α controls a coupling ratio in the linear spectrum but whether a true dynamical scale hierarchy emerges from the nonlinear sector is unsettled and not claimed here.

2.4 Immediate numerical program

The paper now commits to a concrete simulation agenda rather than to a broad list of generic falsification slogans. Three numerical tasks are decisive:

1. **Dispersion and isotropy control.** Measure direction-dependent wave speeds, birefringence, and branch splitting as functions of wavelength relative to the lattice spacing and neighbor stencil. The aim is to determine whether a universal long-wavelength observer sector exists.

2. **Contraction-channel extraction.** Create localized transverse excitation, measure the induced slow in-brane contraction field, and test whether it focuses or accelerates long-wavelength probe packets in the expected weak-field manner.

3. **Localized triplet-mode search.** Construct and evolve confined three-component seeds tied to the axis-aligned cubic channels, and test whether the substrate supports stable baryon-like standing-wave modes with confined internal triplet structure, controlled far-field holonomy, and reproducible size/energy diagnostics.

The contribution of the present paper is therefore a sharpened substrate model plus a concrete route for deciding whether its main emergence claims survive direct computation.

3 Substrate model: lattice microphysics and continuum (IR) effective description

3.1 Ontology and kinematics

The fundamental object of this theory is a *4D brane lattice embedded in a 4D Euclidean ambient with codimension zero*: the lattice is the substrate, and there is no extra perpendicular direction in the ambient. The ambient $\mathbb{R}^4$ is fully symmetric — no preferred direction — and serves only as a stage on which Euclidean distances are measured. All asymmetries that we will encounter (the distinction between time and the three spatial directions, the split between gauge and gravity sectors) are properties of the brane and of the inside observer's frame, not of the ambient.

Let $\Omega \subset \mathbb{R}^4$ be the intrinsic (material) brane domain (either a periodic cell $\Omega = \mathbb{T}^4$ or a bounded hypercube with clamped boundary). The fundamental field is the embedding map

$$\mathbf{X}(x) \in \mathbb{R}^4, \qquad x = (x^1, x^2, x^3, x^4) \in \Omega, \tag{2}$$

whose components $X^A(x)$ $(A = 1, \dots, 4)$ give the brane configuration in the ambient. The four ambient directions are geometrically equivalent. The asymmetry that picks one of them as the *timelike* direction is a property of the brane action $S[\mathbf{X}]$, not of the ambient: the action has Lorentzian sign structure — the standard $T - V$ Lagrangian, recast on the lattice as link-energy terms with opposite sign for the timelike-direction contributions. Globally, cosmological boundary conditions (a Big Bang vertex emanating along one specific ambient direction) fix this choice unambiguously, so every inside observer agrees on which axis is time.

Working spacelike-slice description. For technical work that does not require explicit 4D treatment — the linear-regime dispersion of Section 4, the gauge-sector diagnostics of Section 5, and the localized-mode analysis of Section 7 — it is convenient to slice the brane perpendicular to the timelike direction. Each spacelike slice is parameterised by $x = (x^1, x^2, x^3) \in \Omega_3 \subset \mathbb{R}^3$, with the timelike coordinate $t := x^4$ acting as the slicing parameter. In this description the embedding takes the equivalent form

$$\mathbf{X}(x, t) \in \mathbb{R}^4, \qquad x = (x^1, x^2, x^3) \in \Omega_3, \quad t \in \mathbb{R}, \tag{3}$$

that of a three-dimensional material domain embedded in the 4D ambient. The fourth ambient component, denoted u in this slicing, is the inside observer's chosen timelike component of $\mathbf{X}$; relative to the spacelike slice it plays the role of a transverse "amplitude" direction. With this identification, all the constitutive, kinematic, and gauge-sector developments below carry through verbatim, now understood as describing the inside observer's spacelike slice.

3.2 Discrete lattice substrate (microphysical postulate)

The continuum model is treated as the long-wavelength effective theory of a *4D cubic lattice* substrate. Lattice sites are labeled by $n \in \mathbb{Z}^4$ with spacing a and reference positions $x_n = a\,n$; each site carries an embedded configuration $\mathbf{X}_n \in \mathbb{R}^4$. The discrete brane action is a sum of link-energy contributions with the Lorentzian sign structure of Section 3.1: contributions involving the timelike direction enter with opposite sign from those involving the three spacelike directions.

On a fixed spacelike slice (perpendicular to the timelike direction), sites are labeled by $n \in \mathbb{Z}^3$ with embedded configurations $\mathbf{X}_n(t) \in \mathbb{R}^4$ where t is the slicing parameter. For a spacelike neighbour stencil $\mathcal{N} \subset \mathbb{Z}^3$, the slice-restricted potential energy reads

$$U_{\mathrm{lat}}[\{\mathbf{X}_n\}] = \frac{1}{2} \sum_n \sum_{\delta \in \mathcal{N}} k_\delta \Big(|\mathbf{X}_{n+\delta} - \mathbf{X}_n| - \alpha a |\delta| \Big)^2, \tag{4}$$

with stiffnesses k_δ grouped, for example, into axis-aligned, face-diagonal, and body-diagonal shells. The uniform prestretch parameter α fixes the ratio between the held lattice spacing and the stress-free link length. Link contributions along the timelike direction enter the full brane action with opposite sign and are absorbed into the kinetic term $\frac{1}{2}\rho_m |\partial_t \mathbf{X}|^2$ of the slice Lagrangian (Equation (23) below); their explicit lattice form is developed in Section 3.3.

Geometric nonlinearity already present at the microscopic level. Equation (4) makes an important modeling point explicit. Even if the link energy is Hooke-linear in the scalar extension, its dependence on the node coordinates is nonlinear because the extension is the full Euclidean distance in $\mathbb{R}^4$. This Pythagorean nonlinearity is the microscopic origin of the amplitude–lateral coupling that later appears in the continuum as the induced-metric term $\partial_i u\, \partial_j u$. No additional nonlinear interaction field is required for this effect.

Minimal stencil and lab-frame anisotropy. The canonical realisation of the lattice in this paper is the *6-neighbor axial-only* stencil: each site connects to its six nearest axial neighbours $\pm\hat{e}_i$. Diagonal-shell (face- and body-diagonal) bonds are intentionally absent. On this stencil the long-wavelength dispersion is cubically anisotropic at leading order in $|k|a$: for example, the longitudinal speed along [100] exceeds the longitudinal speed along [111] by a factor $1/\sqrt{(3-2\alpha)/3}$, evaluating to roughly 7.5% at the default operating point $\alpha = 0.2$. This anisotropy is not retuned away; it is the structural feature that the gauge-sector emergence story in Sections 5 and 6 ultimately rests on.

Dual-observer framework: lab anisotropy vs internal isotropy. The lab (outside) observer knows the lattice and resolves direction-dependent wave speeds. The internal (soliton) observer is built from substrate excitations and, by hypothesis, has rods, clocks, and signal cones that all renormalize with the same direction-dependent local wave speed. Their measured speed of light, defined operationally as the ratio of an emergent rod length to an emergent clock period, is therefore conjectured to be direction-independent in their own units. This is a strong assumption: it promotes “operational isotropy” from a passive consistency condition to the load-bearing mechanism by which Lorentz-like kinematics emerge for internal observers despite manifest lab-frame anisotropy. The decisive empirical test is whether two solitons in different orientations report the same effective metric; this is the central numerical objective of Section 4 and is operationalised in the validation roadmap (Sprint 3).

Brillouin zone and geometric-phase diagnostics. The lattice spacing a defines a Brillouin zone $k_i \in [-\pi/a, \pi/a]$. Because Berry curvature is naturally defined over parameter spaces such as $\mathbf{k}$-space eigenbundles, the lattice formulation provides a direct route to Berry-curvature diagnostics on a discretized momentum lattice (see Section 5).

3.3 Explicit discrete 4D brane action

The spacelike-slice description of Section 3.2 treats the timelike direction implicitly, absorbing it into the kinetic term $\frac{1}{2}m|\dot{R}|^2$. For the foundational 4D-in-4D ontology of Section 3.1 it is essential to make the timelike direction an explicit lattice link, placing all four spatial directions on equal footing as far as the link topology is concerned and making the Lorentzian sign structure a single explicit choice. Label each node by a spacelike triple $p = (i, j, k)$ and a timelike index $l \in \mathbb{Z}$, with 4D embedded position $R_p^l \in \mathbb{R}^4$. The six spacelike axial neighbours form the stencil $\mathcal{N}_s = \{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$ and the two temporal neighbours form $\mathcal{N}_t = \{l \pm 1\}$.

The action. The slice-restricted potential is the central-force sum over $\mathcal{N}_s$:

$$V^l = \frac{k_s}{4} \sum_p \sum_{\delta \in \mathcal{N}_s} \Big(|R_{p+\delta}^l - R_p^l| - \alpha a \Big)^2, \tag{5}$$

and the temporal kinetic term on the link $l \to l+1$ is

$$T^{l+\frac{1}{2}} = \sum_p \frac{m}{2} \left(\frac{R_p^{l+1} - R_p^l}{\Delta t} \right)^2. \tag{6}$$

With the Lorentzian sign assignment — spacelike links enter with $-$ (potential), temporal links enter with $+$ (kinetic); this is Assumption A5 of the derivation and is *posited*, not derived: it is the property of the brane action that selects one of the four equivalent ambient directions as timelike (backbone #21) — the explicit discrete action is

$$\boxed{S[R] = \sum_l \Delta t \left(T^{l+\frac{1}{2}} - V^l \right).} \tag{7}$$

This is a sum over $6 + 2$ links per node with a single sign flip for the temporal family.

Euler–Lagrange identity: Verlet is the variational integrator. Varying Equation (7) with respect to an interior node R_p^l (the node appears in $T^{l\pm\frac{1}{2}}$ and V^l) yields

$$m \, \frac{R_p^{l+1} - 2R_p^l + R_p^{l-1}}{\Delta t^2} \;=\; F_p^l \;:=\; -\frac{\partial V^l}{\partial R_p^l} = k_s \sum_{\delta \in \mathcal{N}_s} \Big(|R_{p+\delta}^l - R_p^l| - \alpha a \Big) \, (\widehat{R_{p+\delta}^l - R_p^l}), \tag{8}$$

which is precisely the Störmer–Verlet update $R_p^{l+1} = 2R_p^l - R_p^{l-1} + (\Delta t^2/m)\, F_p^l$. This is the discrete-mechanics result (Marsden–West): the Euler–Lagrange equations of the discrete action S are exactly Störmer–Verlet, and the flow is symplectic. Therefore *Verlet is the discrete variational integrator of the action* (7). The forward initial-value problem (prescribe R^0, R^1, march $l \to l+1$) and the foundational 4D block boundary-value problem (prescribe spacelike-slice data at $l = 0$ and $l = N$, root-find the interior) share *identical local stencil* (8) and differ only in their global boundary conditions.

Long-wavelength cone speeds. Taking the continuum limit with $\rho_m = m/a^3$ and wave vector $\mathbf{k} = k\hat{e}_x$, the Christoffel matrix is diagonal on the axial stencil $\mathcal{N}_s$ (no off-diagonal coupling at any α), giving the lattice light-cone speeds

$$c_L^2 = \frac{k_s \, a^2}{m}, \qquad c_T^2 = (1 - \alpha) \, \frac{k_s \, a^2}{m}. \tag{9}$$

In dimensionless units $k_s = a = m = 1$ these reduce to $c_L = 1$, $c_T = \sqrt{1 - \alpha}$. At the default operating point $\alpha = 0.2$ one obtains $c_T/c_L = \sqrt{0.8} \approx 0.894$, a residual $\sim 10.6\%$ longitudinal–transverse gap that is the leading-order consequence of prestretch; it is not retuned away (backbone #8, #15).

Outlook: foundational block solver. The Lorentzian action (7) is a saddle (unbounded below), so a foundational block solver must root-find $\nabla S = 0$ rather than minimize S. The timelike link is a central-force spring like the three spacelike links, so the substrate is one 4D-isotropic spring lattice parameterized by the temporal rest length r_t: $r_t = 0$ is the linear/Verlet limit used throughout this paper, and $r_t = \alpha\,\beta\,\Delta t$ is the prestressed substrate carrying a timelike geometric quartic. The well-posedness of the two-time boundary-value problem (resolved in the linear regime: a two-past-slice chiral boundary condition has an N-independent, bounded condition number) and the consistency of a single 4D prestress $\alpha_t = \alpha$ are tracked outside the manuscript.

3.4 Induced metric and isotropic reference metric

The embedding induces a (time-dependent) Riemannian metric on the brane:

$$g_{ij}(x,t) := \partial_i \mathbf{X}(x,t) \cdot \partial_j \mathbf{X}(x,t), \qquad i,j = 1,2,3. \tag{10}$$

This definition is invariant under rigid motions $\mathbf{X} \mapsto Q\mathbf{X} + a$ with $Q \in O(4)$ and $a \in \mathbb{R}^4$. Rotational invariance within the brane is obtained by constructing the stored energy density from invariants of g relative to an isotropic reference metric.

We choose a homogeneous, isotropic reference metric

$$g^0_{ij} = \ell_0^2\,\delta_{ij}, \tag{11}$$

where $\ell_0 > 0$ is the continuum analogue of a discrete spring rest length. Define the mixed relative metric

$$\widehat{g} := (g^0)^{-1} g. \tag{12}$$

Any isotropic hyperelastic energy density can be written as $W = W(\widehat{g})$, equivalently as a function of the principal stretches or invariants of $\widehat{g}$.

Continuum analogue of pre-tension. The reference metric g^0 encodes the stress-free geometric scale. If boundary conditions enforce a held ground-state scale $h_\star$ (uniform), then

$$\alpha := \frac{\ell_0}{h_\star} \in (0,1] \tag{13}$$

measures the mismatch between stress-free scale and held scale. For $\alpha < 1$ the brane is uniformly pre-stressed (continuum "pre-tension"). This mismatch controls how strongly transverse gradients backreact on in-brane stresses through the geometric nonlinearity of $g_{ij}(\mathbf{X})$. In the lattice substrate picture (Section 3.2), the same α multiplies the geometric neighbor distances to set spring rest lengths; Equation (13) is the continuum encoding of that uniform prestretch.

3.5 Constitutive law: central-force spring and its continuum expansion

The exact constitutive law. The substrate's stored energy is determined entirely by the central-force spring potential. On the 6-neighbor axial-only stencil $\mathcal{N}_6 = \{\pm\hat{e}_i\}$, the exact slice-restricted potential (Equation (5)) is

$$V = \frac{k_s}{4} \sum_p \sum_{\delta \in \mathcal{N}_6} \Big(|R_{p+\delta} - R_p| - \alpha a \Big)^2. \tag{14}$$

This is the energy the simulation kernel `branesim/core/action.py` computes exactly. It is *frame-indifferent*: it depends only on the Euclidean link length $|R_{p+\delta} - R_p|$, not on rigid motions of the ambient. The induced-metric line element $|\Delta R|^2 = |\partial_i \mathbf{X}\,\mathrm{d}x^i|^2$ is the discrete counterpart of the Riemannian pull-back (10): both measure the physical distance between neighboring material points, confirming that (14) is the natural elastic energy for an embedded lattice.

Exact link decomposition. Writing $E_{\text{link}} = \frac{1}{2}k_s(|\Delta R| - \alpha a)^2$ and splitting into the squared-norm and the Euclidean-norm terms,

$$E_{\text{link}} = \underbrace{\tfrac{1}{2}k_s|\Delta R|^2}_{\text{exactly quadratic}} - \underbrace{k_s\alpha a\,|\Delta R|}_{\text{entire anharmonic sector}} + \tfrac{1}{2}k_s\alpha^2 a^2. \tag{15}$$

The squared-norm term $\frac{1}{2}k_s|\Delta R|^2$ is a degree-2 polynomial in node coordinates and contributes *only* to the quadratic (linear-elasticity) sector. Every cubic and higher term in the energy descends from the Euclidean-norm term $-k_s\alpha a\,|\Delta R|$, which carries an explicit factor α. Consequently, *all anharmonic physics vanishes as* $\alpha \to 0$ (backbone #17).

Continuum expansion: $W = W_2 + W_4 + \cdots$ In the long-wavelength regime ($a|k| \ll 1$, $|\nabla u| \ll 1$) the exact spring energy coarse-grains to a gradient expansion $W = W_2 + W_4 + \cdots$ derived in `paper/derivations/spring_constitutive_continuum_limit.md`.

Leading term W_2 (linear elasticity). Using the quadratic link strain $(s_\delta^2)^{(2)} = \alpha(\hat\delta\cdot\Delta u)^2 + (1-\alpha)|\Delta u|^2$ and the axial tensor sums $T^{(2)}_{ab} = 2\delta_{ab}$, $T^{(4)}_{abcd} = 2Q_{abcd}$ (where $Q_{abcd} = 1$ iff $a = b = c = d$), the acoustic tensor on $\mathcal{N}_6$ evaluates to

$$A_{abcd} = \frac{k_s}{a}\big[\alpha\,Q_{abcd} + (1-\alpha)\,\delta_{ac}\delta_{bd}\big]. \tag{16}$$

The Christoffel matrix along $k = k\hat e_1$ is diagonal, giving

$$c_L^2 = \frac{k_s a^2}{m}, \qquad c_T^2 = (1-\alpha)\,\frac{k_s a^2}{m}, \tag{17}$$

in exact agreement with the lattice closed-form dispersion of `branesim/core/conventions.py` (cross-checked at every α). The effective Lamé parameters on the 6-neighbor stencil are

$$\mu(\alpha) = \frac{(1-\alpha)\,k_s}{a}, \qquad \lambda = C_{1122} = 0. \tag{18}$$

The vanishing of λ and the off-diagonal entries of A_{abcd} confirm that the 6-neighbor stencil is structurally *cubically anisotropic* ($C_{1111} - C_{1122} - 2C_{1212} = (2\alpha-1)k_s/a \neq 0$ for $\alpha \neq \frac{1}{2}$): this is a feature, not a bug — it is the structural source of the gauge sector (backbone #8, #15, #16, #19).

Cubic term $W_3 = 0$. On purely longitudinal modes ($\Delta u_\perp = 0$) the link length is $|\Delta R| = |a + \Delta u_\parallel|$, which is exactly Hookean — no odd anharmonic terms. On purely transverse modes the displacement $q \equiv |\Delta u_\perp|$ appears only through q^2, so $W_3 = 0$ by parity. *The geometric nonlinearity is intrinsically transverse.*

Quartic term W_4 (soliton stabilizer). Expanding the Euclidean-norm term on the pure-transverse slice ($\Delta u_\parallel = 0$, $q = |\Delta u_\perp|$):

$$|\Delta R| = \sqrt{a^2 + q^2} = a + \frac{q^2}{2a} - \frac{q^4}{8a^3} + \cdots, \tag{19}$$

and collecting the quartic contribution from $-k_s\alpha a\,|\Delta R| \supset +(k_s\alpha/8a^2)q^4$, the continuum coarse-graining gives

$$\boxed{W_4 = \frac{k_s\alpha}{8a}\,|\partial u_\perp|^4,} \tag{20}$$

with coefficient $\propto \alpha$ that *vanishes at* $\alpha \to 0$ and is positive (geometric hardening, $Q > 0$, backbone #17). In 4D the transverse component includes both the lateral and the timelike displacement channels, producing the lateral–amplitude geometric quartic $(k_s\alpha/4a)|\partial u^{\text{lat}}|^2(\partial u^{X^4})^2$ that stabilizes solitons against Derrick (uniform-rescaling) collapse (backbone #17). This is the standard Skyrme balance: $W_2 \sim \lambda^{-1}$ and $W_4 \sim \lambda^{+1}$ under $u(x) \mapsto u(\lambda x)$ in 3D — here λ denotes the Derrick rescaling factor, not the Lamé parameter (which vanishes on this stencil) — with an independent UV cutoff at the lattice spacing a.

Green–Lagrange strain and the StVK quadratic-order proxy. For analytic comparison with the textbook literature it is useful to introduce the Green–Lagrange strain tensor

$$E := \tfrac{1}{2}(\widehat{g} - I) \qquad \text{(with } I \text{ the } 3\times 3 \text{ identity)}, \tag{21}$$

and the St. Venant–Kirchhoff (StVK) stored energy density

$$W_{\mathrm{StVK}}(E) = \mu\,\mathrm{tr}(E^2) + \frac{\lambda}{2}\,(\mathrm{tr}E)^2. \tag{22}$$

StVK is a quadratic-order proxy for the central-force spring, not the substrate's constitutive law. At quadratic order in ∇u, StVK reproduces the same wave speeds (17) and the same μ, λ as the spring (18): the two descriptions are indistinguishable in the linear-elasticity regime of Sections 4 and 5. At quartic order, however, they diverge critically: the spring's geometric quartic (20) has coefficient $\propto k_s\alpha/a$, whereas a naive StVK identification locks the quartic to $\mu \propto (1-\alpha)$ and *inverts* the α-scaling. *Quantitative soliton sizing (Section 7) must use the spring law* (14)–(20), *not StVK.*

On the 6-neighbor stencil the vanishing of λ (Equation (18)) gives Poisson ratio $\nu = 0$ structurally.

Compression non-coercivity. The central-force spring (14) is not polyconvex nor coercive in compression: as a link collapses ($|R_{p+\delta} - R_p| \to 0$), the pair energy $\frac{1}{2}k_s(\alpha a)^2$ remains bounded rather than diverging. The lattice spacing a masks this in practice (configurations cannot compress below the grid), but the continuum limit exposes it. For the linear and moderate-strain results of Sections 4 and 5 this is immaterial; for the strongly localized modes of Section 7 it must be controlled — either by verifying that the relevant configurations remain non-inverting, or by substituting a polyconvex hyperelastic energy (e.g. neo-Hookean, Mooney–Rivlin, Ogden-type) within the same induced-metric framework, which leaves the geometric mechanism of transverse–lateral backreaction unchanged. The geometric quartic (20) that stabilizes solitons against Derrick collapse is a distinct safeguard and does not by itself remove this local compression instability.

Spacelike-slice versus 4D placement. The strain (21) is a spacelike-slice object: E is 3×3 and is referred to the Euclidean reference metric of the slice. Its placement in the foundational 4D action (Section 3.3) follows the single timelike-link model: the spring potential V (14) covers the spacelike links and the timelike link enters with the opposite sign, reducing in the $r_t = 0$ limit to a separate kinetic term so that $S = \int(T - V)$ is ordinary elastodynamics. The Lorentzian signature resides in the *sign structure of the action* (Section 3.1), not in the ambient or induced metric: the ambient is Euclidean, so $g_{\mu\nu} = \partial_\mu \mathbf{X}\cdot\partial_\nu\mathbf{X}$ is positive-definite and the timelike direction is distinguished only by the opposite sign with which its contribution enters S.

3.6 Lagrangian, action, and equations of motion

Let $\rho_m > 0$ be the brane mass density (kg/m^3). The Lagrangian density is

$$\mathcal{L}(\mathbf{X}, \partial_t\mathbf{X}, \nabla\mathbf{X}) = \frac{\rho_m}{2}\,|\partial_t\mathbf{X}|^2 - W\Big(g(\mathbf{X}); g^0\Big). \tag{23}$$

The action is

$$S[\mathbf{X}] = \int_{\mathbb{R}} \mathrm{d}t \int_{\Omega} \mathrm{d}^3x\,\mathcal{L}. \tag{24}$$

Define the first Piola stress (as a momentum conjugate to $\partial_i X^A$)

$$P_A^{\ i} := \frac{\partial W}{\partial(\partial_i X^A)}. \tag{25}$$

For energies depending on g_{ij} one can write $P_A^{\ i} = 2S^{ij}\partial_j X^A$ with a symmetric second Piola-type tensor $S^{ij} = \partial W/\partial g_{ij}$. The Euler–Lagrange equations take divergence form

$$\rho_m\, \partial_{tt} X^A = \partial_i P_A^{\ i}, \qquad A = 1, \ldots, 4, \tag{26}$$

showing explicitly that isotropy (absence of built-in preferred directions) follows from writing W in terms of invariants of the induced metric relative to the reference metric g^0.

3.7 Spherical material coordinates (coordinate change only)

Important: this is only a coordinate change. All dynamical statements remain those of the coordinate-free formulation $\mathbf{X} : \Omega \times \mathbb{R} \to \mathbb{R}^4$ with induced metric $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$. We introduce spherical coordinates on the *intrinsic/material* domain as a convenient chart for localized excitations; this does *not* constrain the embedding $\mathbf{X}$.

Let (r, θ, φ) be the standard spherical chart on (a subset of) $\mathbb{R}^3$, with Jacobian $J(r, \theta, \varphi) = r^2 \sin\theta$. Writing the intrinsic coordinate as $q^a \in \{r, \theta, \varphi\}$ and denoting derivatives in this chart by ∂_a, the equations of motion become

$$\rho_m\, \partial_{tt} X^A = \frac{1}{J(q)}\, \partial_a\Big(J(q)\, P_A^{\ a}\Big), \qquad A = 1, \ldots, 4, \tag{27}$$

which is exactly the Cartesian divergence form (26) written with the standard Jacobian factor.

3.8 Linearization and polarized branches

Linearize around a homogeneous background embedding $\mathbf{X}_\star(x) = (x, 0)$ and small displacements $\mathbf{u}(x,t) = \mathbf{X}(x,t) - \mathbf{X}_\star(x)$. Under standard regularity assumptions on W, one obtains a linear wave system with (in general) multiple polarization branches, including transverse and longitudinal modes. For later use, we emphasize two generic features: (i) different branches may have different characteristic speeds (and may become dispersive at shorter wavelengths), and (ii) for $\alpha < 1$ the pre-stress couples transverse excursions (in the embedding) back into in-brane stresses through the induced metric (10). These branches provide the substrate carriers for the geometric-phase and soliton constructions developed below.

3.9 Transverse–lateral coupling and an emergent "gravity" channel

A distinctive feature of an embedded elastic medium is that transverse gradients change intrinsic distances. To make this explicit without removing in-brane dynamics, we use a near-identity decomposition around the homogeneous background embedding $\mathbf{X}_\star(x) = (h_\star x, 0)$:

$$\mathbf{X}(x,t) = \big(h_\star x^1 + \xi^1(x,t),\ h_\star x^2 + \xi^2(x,t),\ h_\star x^3 + \xi^3(x,t),\ u(x,t)\big), \qquad \vec{\xi}(x,t) \in \mathbb{R}^3. \tag{28}$$

The induced metric is then

$$\begin{aligned} g_{ij} &= \partial_i \mathbf{X} \cdot \partial_j \mathbf{X} \\ &= (h_\star \delta_i^a + \partial_i \xi^a)(h_\star \delta_j^a + \partial_j \xi^a) + \partial_i u\, \partial_j u \\ &= h_\star^2 \delta_{ij} + h_\star(\partial_i \xi_j + \partial_j \xi_i) + \partial_i \vec{\xi} \cdot \partial_j \vec{\xi} + \partial_i u\, \partial_j u. \end{aligned} \tag{29}$$

Setting $\vec{\xi} \equiv 0$ recovers the familiar graph picture used only for intuition; the "gravity" channel, however, relies on the allowed in-brane relaxation $\vec{\xi} \neq 0$ sourced by the quadratic term $\partial_i u\, \partial_j u$.

In the small-slope regime, the in-brane strain has the schematic form

$$\varepsilon_{ij} \approx \tfrac{1}{2}(\partial_i \xi_j + \partial_j \xi_i) + \tfrac{1}{2}\, \partial_i u\, \partial_j u, \tag{30}$$

so the quasi-static balance $\partial_i \sigma_{ij} \approx 0$ with $\sigma_{ij} = \lambda\, \mathrm{tr}(\varepsilon)\delta_{ij} + 2\mu\, \varepsilon_{ij}$ makes it explicit that the term $\partial_i u\, \partial_j u$ acts as a forcing/eigenstrain-like source for the slow in-brane field $\vec{\xi}$.

Pythagorean lengthening and lateral contraction (informal picture). For a small material displacement $\mathrm{d}x$, the ambient line element of the graph embedding satisfies

$$\mathrm{d}s^2 = h_\star^2\,|\mathrm{d}x|^2 + (\nabla u \cdot \mathrm{d}x)^2. \tag{31}$$

Thus any region with nonzero slope ∇u necessarily *lengthens* intrinsic distances compared to the flat configuration. In a discrete spring-lattice realization this is the familiar Pythagorean effect $|\Delta\mathbf{X}| = \sqrt{|\Delta x|^2 + (\Delta u)^2}$: transverse variation increases spring length even when lateral coordinates are unchanged. In the actual dynamics the in-brane displacement $\vec{\xi}$ is not constrained; under pre-tension ($\alpha < 1$) the additional metric contribution $\partial_i u\,\partial_j u$ acts as an effective source in the in-brane equilibrium equations, producing a compensating contraction/dilation pattern rather than a purely local increase of tensile stress. If the medium is pre-tensioned ($\alpha < 1$), the additional length is paid for by increased tensile stress, which produces net lateral forces that pull material points *toward* the region of transverse excitation. For a localized bump or rotating transverse pattern, the surrounding brane therefore develops an inward bias (contraction) rather than an outward pressure.

In this qualitative sense, the "gravitational mass" of an excitation is tied to the elastic energy stored in transverse gradients (and, more generally, to embedded deformation energy), while the "gravitational field" is the slowly varying in-brane strain/dilation pattern sourced by that energy. The sign is fixed by geometry: increasing transverse arc length increases elastic energy and therefore increases tension, which pulls laterally inward.

With homogeneous pre-stress ($\alpha < 1$), this geometrically induced strain backreacts on the in-brane stress field through the constitutive law, effectively coupling transverse excitation energy into slow modulations of the in-brane configuration.

Unification across all four directions. The Pythagorean argument above is direction-blind: in the underlying 4D ontology (Section 3.1) any displacement perpendicular to a link stretches that link, regardless of which ambient direction is perpendicular. The slice-projected description above treats $u = X^4$ specially because the inside observer has already chosen X^4 as the timelike direction. The same mechanism, applied symmetrically across the four directions, predicts that displacements in the three *spacelike* directions (i.e. in-brane kinetic energy / spatial motion) also lengthen the link along the timelike direction — which is gravitational *time dilation* sourced by stress–energy. Length contraction (timelike-direction excitation stretches spacelike links) and time dilation (spacelike-direction excitation stretches the timelike link) are therefore two faces of one rule. The slice-restricted analysis of this subsection captures the first face; a full 4D treatment of the brane action would express both within one expression.

This provides a concrete mechanism for a long-range channel often described informally as "gravity" in the present program: localized transverse gradients can act as sources of slowly varying background strain (contraction/dilation), and linear wave packets propagating on top of that background experience branch-dependent effective coefficients (cf. Section 4). In a weak-field, slowly varying regime one may therefore seek a Newtonian-limit reduction in which a scalar potential built from coarse-grained strain (or elastic energy density) obeys a Poisson-type equation. Deriving such a reduction quantitatively (and relating the resulting coupling constant to G) is an open problem of the present theory.

4 Emergent relativity and effective Lorentz symmetry

The substrate carries Lorentzian sign structure only at the level of the brane *action* (Section 3.1); the ambient $\mathbb{R}^4$ in which the brane is embedded is symmetric Euclidean, and on the spacelike-slice working description t enters as the slicing parameter rather than as a metric direction. A Lorentz *metric* on excitations is therefore not postulated — it is an emergent property of the linear wave regime as seen by an inside observer. This section makes precise the sense in which Lorentz kinematics emerge as a kinematic symmetry of the dominant dispersion branch that controls information transport.

4.1 Linear-wave cone and an effective light speed

Linearizing Equation (26) about a homogeneous background yields a multi-branch wave system. For isotropic elastic media, one expects (at long wavelengths) transverse and longitudinal branches. The characteristic speeds $c_L^2 = k_s a^2/m$ and $c_T^2 = (1-\alpha)\, k_s a^2/m$ are derived in Section 3 from two independent starting points — the closed-form dispersion of the discrete action (Equation (9)) and the acoustic-tensor eigenvalues of the W_2 continuum limit (Equation (17)) — and their agreement there is a self-consistency check across levels of description. For each polarization branch b one has the long-wavelength dispersion

$$\omega_b^2(\mathbf{k}) \;\approx\; c_b^2\, |\mathbf{k}|^2 \quad \text{for } |\mathbf{k}| \to 0, \tag{32}$$

which defines an effective wave cone for that branch.

In regimes where a single branch dominates transport and dispersion is negligible, the wave equation is approximately invariant under the standard Lorentz group at speed c, with boost parameter

$$\gamma = \left(1 - v^2/c^2\right)^{-1/2}, \tag{33}$$

as an effective symmetry of the linear propagation sector, not as a microscopic postulate.

Dual-observer reading: the only Lorentz observer is the internal one. On the 6-neighbor axial-only lattice (Section 3.2) the long-wavelength wave speed is direction-dependent at the lab-frame level — for example $c_L([100])/c_L([111]) \approx 1.075$ at the default operating point $\alpha = 0.2$. The relativistic effective description (33) is therefore not a statement about the lab frame; it is a conjecture about *internal* observers, whose rods, clocks, and signal cones are substrate-emergent and renormalize with the same direction-dependent speed (see Section 3.2 for the full mechanism). Without observer universality the emergent Lorentz statement collapses to "a particular branch propagates with one speed along each direction" and is not a symmetry of physics for any actual observer.

4.2 Analogue-metric viewpoint

An equivalent viewpoint, common in analogue-gravity systems, is that linear excitations experience an effective metric structure with interval

$$ds^2 = -c^2 dt^2 + g_{ij}^{(\mathrm{eff})}(x,t)\, dx^i\, dx^j, \tag{34}$$

where $g_{ij}^{(\mathrm{eff})}$ depends on the background configuration about which one linearizes. In the present model there are two natural geometric objects: the induced brane metric $g_{ij}(\mathbf{X})$ and the branch-dependent effective coefficients of the linearized operator. In particular, slowly varying transverse profiles $u = X^4$ modify intrinsic distances already at the level of the induced metric (Equation (29) and section 3.9), and can therefore be read as a background that focuses or defocuses wave transport in close analogy to a weak gravitational field. Both encode how background deformations can bend or modulate wave propagation, in close analogy to analogue-gravity constructions [7, 50].

4.3 Regime of validity and expected deviations

Effective Lorentz symmetry is expected to break when: (i) multiple branches contribute comparably (mode mixing), (ii) dispersion becomes significant at shorter wavelengths, or (iii) geometric/material nonlinearity becomes important at larger amplitudes. In the present theory these deviations are not treated as flaws; they are structural consequences of a substrate whose timelike direction is selected by the brane action rather than imposed as a metric primitive, and they therefore delimit the domain where a relativistic effective description is expected to apply.

5 Geometric phase, Berry/Wilczek–Zee connections, and gauge structure

5.1 Per-wavepacket band isolation and polarization transport

The complex-envelope description is applied per band-isolated excitation. Each wavepacket is treated as narrowband around its own local carrier angular frequency ω_0, determined by the linearized carrier operator about the slowly varying background at the wavepacket's location; no globally coherent universe-wide carrier is assumed. A band-isolated wavepacket can be written locally as

$$\mathbf{u}(x,t) \;\approx\; \mathrm{Re}\Big\{a(x,t)\,\boldsymbol{\epsilon}(x,t)\,e^{\mathrm{i}\theta(x,t)}\Big\}, \qquad \theta(x,t) = \phi(x,t) - \omega_0 t, \tag{35}$$

where $a(x,t) \geq 0$ varies slowly on the envelope scale, $\boldsymbol{\epsilon}(x,t) \in \mathbb{C}^N$ is a normalized polarization state ($\boldsymbol{\epsilon}^\dagger\boldsymbol{\epsilon} = 1$), and ω_0 is the wavepacket's local carrier angular frequency. The internal dimension N counts the relevant components (e.g. embedding-direction components and/or a coarse-grained mode basis).

A central point is that polarization transport is not unique: the same physical state can be represented in many local phase conventions. Geometric phase formalizes this redundancy as a connection on a bundle of polarization states.

5.2 Rank-1 Berry connection (U(1))

For a single isolated polarization mode, represent the local state by a normalized vector $|u(x)\rangle \in \mathbb{C}^N$ that depends smoothly on spacetime position x^μ (here $x^\mu = (c\,t, x^1, x^2, x^3)$ built from the intrinsic coordinates and a chosen characteristic speed c of the carrier branch). Define the Berry connection

$$a_\mu(x) := \mathrm{i}\,\langle u(x)\,|\,\partial_\mu u(x)\rangle. \tag{36}$$

Under a local phase choice $|u(x)\rangle \mapsto e^{-\mathrm{i}\chi(x)}|u(x)\rangle$ one finds

$$a_\mu \mapsto a_\mu + \partial_\mu\chi, \tag{37}$$

so only gauge-invariant quantities carry physical meaning. The Berry curvature is the field strength

$$f_{\mu\nu} := \partial_\mu a_\nu - \partial_\nu a_\mu, \tag{38}$$

and the holonomy (geometric phase) around a closed loop Γ is

$$\gamma_\Gamma = \oint_\Gamma a_\mu\,\mathrm{d}x^\mu = \int_{\Sigma:\,\partial\Sigma=\Gamma} f_{\mu\nu}\,\mathrm{d}\Sigma^{\mu\nu}. \tag{39}$$

Electromagnetic reading: four-potential and field tensor. In the $U(1)$ trace sector (Section 5.6) the Berry connection (36) *is* the electromagnetic four-potential, $A_\mu \equiv a_\mu$, and its curvature (38) *is* the field tensor $F_{\mu\nu} \equiv f_{\mu\nu}$, with $E_i = F_{0i}$ (the time–space part) and $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$ (the space–space part): the electric and magnetic fields are the two faces of how the carrier's worldtube phase fails to commute between the timelike and spacelike directions. Two consequences are then *automatic*, requiring no field dynamics: the gauge freedom (37) is exactly the unobservable choice of carrier phase reference, and the homogeneous Maxwell equations are the Bianchi identity $\partial_{[\lambda}F_{\mu\nu]} = 0$, which holds identically because $F = dA$. A *nonzero* field ($F \neq 0$) requires the phase to be non-integrable — the polarization state $|u\rangle$ must genuinely rotate within $\mathbb{C}^N$ ($N \geq 2$), not merely carry an overall phase, since a pure gradient $A_\mu = \partial_\mu \chi$ is pure gauge. Because the index μ runs over the 4D worldtube, A_μ transforms as a Lorentz four-vector only insofar as the emergent light-cone is isotropic (Section 4). What this kinematics does *not* fix is the field's *dynamics*: the inhomogeneous equations $d{\star}F = J$ hold only if the substrate furnishes the Maxwell action $-\frac{1}{4}\int F \wedge {\star}F$ (with its emergent Hodge star and coupling), a separate derivation not established here.

5.3 What the geometric phase is "of": eigenmode bundles and adiabaticity

The connection above is not attached to an arbitrary decomposition of a real field into amplitude and phase. In the present setting one should choose $|u(x)\rangle$ as a (locally defined) eigenvector of the linearized carrier operator about the slowly varying background configuration at x. As x varies, these eigenvectors form an eigenbundle, and the Berry/Wilczek–Zee connection is the natural connection on that bundle; this is the same eigenbundle picture that underlies Berry-phase effects in crystalline electronic systems [52], here applied to the polarization eigenbundle of the mechanical carrier rather than to electronic Bloch bands.

A clean rank-1 ($U(1)$) reduction requires that the dynamics remains in a single isolated band. If $\Delta\omega_{\mathrm{gap}}(x)$ is the local separation to the nearest competing band and $\Omega_{\mathrm{mix}}(x)$ is the rate of non-adiabatic mode conversion induced by background gradients, then the adiabatic condition

$$\Omega_{\mathrm{mix}}(x) \ll \Delta\omega_{\mathrm{gap}}(x) \tag{40}$$

should hold on the envelope scale. When this fails, a measured "Berry signal" typically reflects mode mixing or beating, and the correct reduction is to retain the relevant multi-dimensional subspace (the Wilczek–Zee description). In the brane model, the pre-stress parameter α and the wavelength regime control branch separation and therefore how well an Abelian gauge description can apply.

5.4 Diagnostic protocol and artifact controls (numerical extraction)

The Berry/Wilczek–Zee connection is meaningful only when it is tied to a specific eigenbundle (or transported subspace) and when the reported observables are gauge-invariant. For numerical experiments on the lattice substrate, the following protocol provides a concrete, reproducible procedure to extract holonomy/curvature and to rule out common artifacts.

Protocol

1. **Carrier isolation / demodulation.** Choose a carrier frequency ω_0 and isolate a narrowband sector (e.g. by band-pass filtering in time or by short-time Fourier methods). Work with a complex envelope representation consistent with Equation (35).

2. **Define the instantaneous polarization object.** At each spacetime point (or each sample along a loop), define either (i) a normalized rank-1 state $|u(x)\rangle$ for an isolated band, or (ii) an n-dimensional subspace projector $P(x)$ (or an orthonormal frame $\mathbf{E}(x)$) for a near-degenerate sector. The definition should be tied to the *local linearized carrier problem* about the slowly varying background at x (cf. Section 5.3).

3. **Measure the adiabaticity/gap indicators.** Estimate a local band gap $\Delta\omega_{\mathrm{gap}}(x)$ (separation to the nearest competing band) and a mode-leakage proxy along the transport path (e.g. growth of energy outside the chosen band/subspace). Use these to empirically validate the adiabatic criterion Equation (40).

4. **Holonomy from overlaps (gauge-invariant).** For a discretized loop $\Gamma : x_0 \to x_1 \to \cdots \to x_m = x_0$:
   - *Abelian (rank-1):* define links $U_k = \frac{\langle u(x_k) \,|\, u(x_{k+1})\rangle}{|\langle u(x_k) \,|\, u(x_{k+1})\rangle|} \in U(1)$ and report the loop phase $\gamma_\Gamma = \arg\big(\prod_{k=0}^{m-1} U_k\big)$.
   - *Non-Abelian (rank-n):* define overlap matrices $M_k = \mathbf{E}(x_k)^\dagger \mathbf{E}(x_{k+1})$ and extract their unitary part (e.g. by polar decomposition) as $U_k \in U(n)$. Report the holonomy matrix $U_\Gamma = \prod_{k=0}^{m-1} U_k$ and gauge-invariant summaries such as $\mathrm{spec}(U_\Gamma)$ and $\mathrm{tr}(U_\Gamma)$.

5. **Curvature by small loops / plaquettes.** Approximate curvature by evaluating holonomy around small loops and dividing by enclosed area (real-space), or use the link-variable Brillouin-zone construction in Section 5.7 (momentum-space).

Artifact controls (required for credibility).

- **Gauge randomization test:** apply arbitrary local phase choices (rank-1) or local $U(n)$ frame rotations (rank-n) and confirm that reported invariants (e.g. γ_Γ, $\mathrm{spec}(U_\Gamma)$) do not change.

- **Resolution/convergence:** verify stability of invariants under refinement of Δt, lattice spacing, and loop sampling density.

- **Adiabatic breakdown demonstration:** intentionally violate Equation (40) and show that the extracted holonomy becomes non-robust (strong dependence on sampling/path or visible subspace leakage), distinguishing geometry from beating.

5.5 Wilczek–Zee connection (non-Abelian)

If the relevant polarization sector is an n-dimensional near-degenerate subspace, choose an orthonormal frame $\mathbf{E}(x) \in \mathbb{C}^{N\times n}$ with columns $|u_a(x)\rangle$ spanning the subspace, so that $\mathbf{E}^\dagger \mathbf{E} = I_n$. Define the Wilczek–Zee connection

$$\mathcal{A}_\mu(x) := \mathrm{i}\,\mathbf{E}(x)^\dagger\,\partial_\mu \mathbf{E}(x) \in \mathfrak{u}(n). \tag{41}$$

A change of local frame $\mathbf{E} \mapsto \mathbf{E}\,U(x)$ with $U(x) \in U(n)$ yields the non-Abelian gauge law

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U + \mathrm{i}\,U^\dagger \partial_\mu U, \tag{42}$$

and the curvature (field strength) is

$$\mathcal{F}_{\mu\nu} := \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - \mathrm{i}\,[\mathcal{A}_\mu, \mathcal{A}_\nu]. \tag{43}$$

5.6 Axis-aligned triplet and prestress-driven $U(1)^3 \to U(3)$ degeneracy

On the 6-neighbor axial-only cubic lattice (Section 3.2) it is natural to organize the per-wavepacket complex envelope as a three-component triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$ aligned with the lattice axes. The role of the prestress parameter α (derivation convention: α = rest_length/spacing, so $\alpha = 1$ is no prestress and $\alpha = 0$ is maximum prestress) is to control the *eigenvalue spread* of the 3-channel Christoffel matrix $D(k)$ — not to introduce off-diagonal coupling between the axis components, which is absent on this lattice at every α.

Non-degenerate limit ($\alpha \to 1$, no prestress). The lattice energy is purely Hookean longitudinal, and $D(k)$ has maximally split eigenvalues with the transverse modes at zero frequency. The three axis-aligned channels are dynamically decoupled: each component supports an independent local phase choice and an independent rank-1 Berry connection $a_\mu^{(x)}, a_\mu^{(y)}, a_\mu^{(z)}$. In this limit, it is meaningful (and gauge-invariant) to speak of "the Berry phase of an axis." The carrier-triplet gauge group is closest to $U(1)^3$.

Degenerate limit ($\alpha \to 0$, maximum prestress). The $(1-\alpha)|\Delta u|^2$ geometric-stiffness term dominates and contributes equal stiffness to all components on every bond. $D(k) \propto I$ becomes proportional to the identity at every k, and the three lateral channels are fully degenerate. The transported subspace is genuinely 3-dimensional and the appropriate adiabatic-transport object is a Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ as in Equation (41), decomposable into a $U(1)$ trace part and a traceless $SU(3)$ part (Equation (49)). At the project's default operating point $\alpha = 0.2$ the lattice sits close to this degenerate limit.

What the microscopic anisotropy actually provides. The $U(3)$ gauge group on a 3-dimensional complex envelope is a generic representation-theoretic fact about $\mathbb{C}^3$ and does not require the lattice to be anisotropic. What the cubic anisotropy does provide is the *physically meaningful decomposition*: a preferred 3-dim subspace (the lateral triplet, distinct from the embedding-amplitude direction X^4), a preferred axis-aligned basis within it (which makes the eight $SU(3)$ traceless generators carry operational meaning, e.g. as "colour" labels on solitons that occupy specific axis-aligned channels), and a natural identification of the $U(1)$ trace with the electromagnetism-like sector. Without the cubic anisotropy of the lab frame, no preferred basis would exist and the $SU(3)$ generators would be arbitrary linear combinations with no operational handle. This is the load-bearing role of microscopic anisotropy in the gauge-emergence story; see `BACKBONE.md` #19 for the corresponding sector mapping. That material anisotropy can carry genuine non-Abelian gauge structure in real space is not unique to the present construction: Chen et al. [11] show that the off-diagonal permittivity/permeability of a wide class of anisotropic optical media act as a pair of spin-dependent vector potentials, realizing a synthetic $SU(2)$ gauge field in a homogeneous medium with an extractable Wilson loop and Zitterbewegung-like trajectories. The present claim is the analogous statement for the mechanical polarization eigenbundle of the cubic brane, extended to the triplet sector.

5.7 Brillouin-zone link-variable method — and why the gauge structure lives in (x, t)

Where the gauge structure lives. The emergent gauge structure of this model is *not* Brillouin-zone band topology. The dynamical matrix $D(\mathbf{k})$ is the Hessian of a real-valued spring energy, hence real and symmetric for every $\mathbf{k}$; real symmetric matrices admit real eigenvectors, and a real eigenvector bundle over a simply-connected region of the Brillouin zone is topologically trivial — every plaquette holonomy reduces to the identity and every continuous Berry/Wilczek–Zee curvature vanishes identically. (We have verified this numerically across a full α sweep and over the entire (k_x, k_y) plane; the holonomy is identity to machine precision at every plaquette and every α.) The physical gauge connection instead lives over *physical spacetime* (x, t): it is the holonomy of the complex carrier envelope $\Psi \in \mathbb{C}^3$ (Section 5.6), transported around loops in real space and time, with the $U(1)$ trace read as the electromagnetic sector. Bloch's theorem and the Brillouin zone enter only to supply the *carrier band* and the narrowband isolation of a wavepacket about its carrier $\mathbf{k}_0$; they are not the base space of the gauge connection.

The reason the base space is (x, t) and not a spatial slice is that the complex structure itself originates in the *timelike* direction: the $U(1)$ phase is the rotation of the real carrier field as one advances along the time axis, the two real degrees of freedom that span $\mathbb{C}$ being the field and its time-quadrature ($\Psi \approx \xi + \frac{i}{\omega_0}\dot{\xi}$; here "$i$" is the quarter-period time shift, i.e. the time-evolution generator $i\partial_t \Psi = H_{\text{eff}}\Psi$). A single spacelike slice is a real snapshot and carries no phase; the $U(1)$ exists only across the time extent of the worldtube — equivalently, two adjacent time slices are the minimum data that fix it.

The link-variable construction below is accordingly a *template diagnostic* that returns nonzero curvature only once an intrinsically complex effective operator is supplied (the envelope's $i\partial_t \Psi = H_{\text{eff}}\Psi$); applied to the bare real $D(\mathbf{k})$ alone it yields identically zero and does not separate $U(1)^3$ from $U(3)$.

For a lattice substrate, Berry curvature can be computed directly on the discretized Brillouin zone using link variables and plaquettes, closely analogous to lattice-QCD Berry-curvature constructions [21, 41, 53]. Let $|u(\mathbf{k})\rangle$ be a normalized eigenvector (or $\mathbf{E}(\mathbf{k})$ a frame of a degenerate subspace) on a momentum lattice with spacing $\tilde{a}$. In the Abelian case one defines links

$$U_j(\mathbf{k}) = \frac{\langle u(\mathbf{k}) \,|\, u(\mathbf{k} + \tilde{\jmath})\rangle}{|\langle u(\mathbf{k}) \,|\, u(\mathbf{k} + \tilde{\jmath})\rangle|}, \tag{44}$$

and a plaquette

$$U_{ij}(\mathbf{k}) = U_i(\mathbf{k}) U_j(\mathbf{k} + \tilde{\imath}) U_i^\dagger(\mathbf{k} + \tilde{\jmath}) U_j^\dagger(\mathbf{k}), \tag{45}$$

whose argument approximates the integrated Berry curvature over the plaquette. In the non-Abelian case one uses overlap matrices between frames and projects to $U(n)$ links, then forms the same plaquette structure to obtain a discretized non-Abelian curvature. This is a natural diagnostic bridge between the lattice-substrate hypothesis and QCD-motivated non-Abelian geometry.

5.8 Spinorial holonomy and the 4π aspect

For $n = 2$ (a two-level polarization subspace), holonomy lives naturally in a double cover of $SO(3)$. A closed loop in physical space that induces a 2π rotation of an internal polarization frame can therefore produce a sign change of the transported state, while a 4π loop returns it to itself. In the present theory this spinorial 4π aspect is attributed to Berry/Wilczek–Zee holonomy of an internal polarization bundle, and is treated as part of the particle-sector structure developed in Section 7.

6 From mode-frame transport to gauge-covariant envelope dynamics

6.1 Multiscale ansatz and retained polarization subspace

Using the per-wavepacket band isolation of Section 5.1, write the substrate displacement as a slowly varying complex envelope multiplied by a local mode frame:

$$\mathbf{u}(x,t) \;\approx\; \mathrm{Re}\left[\mathrm{e}^{-\mathrm{i}\omega_0 t}\sum_{a=1}^{n}\Psi^a(x,t)\,\mathbf{e}_a(x,t)\right], \tag{46}$$

where $\Psi(x,t)\in\mathbb{C}^n$ varies slowly compared to ω_0^{-1} and $\{\mathbf{e}_a(x,t)\}_{a=1..n}$ is an orthonormal frame spanning the retained polarization subspace. The frame is not unique: $\mathbf{e}_a \mapsto \sum_b \mathbf{e}_b\,U_{ba}(x)$ with $U(x)\in U(n)$ represents the same physical subspace.

Projecting the full substrate dynamics onto the retained subspace produces, generically, a gauge-covariant derivative acting on Ψ:

$$\mathcal{D}_\mu\Psi = (\partial_\mu + \mathrm{i}\,\mathcal{A}_\mu)\,\Psi, \tag{47}$$

where $\mathcal{A}_\mu$ is the Berry (rank-1) or Wilczek–Zee (non-Abelian) connection defined in Section 5. The gauge transformation $\mathbf{E}\mapsto\mathbf{E}U$ induces $\Psi\mapsto U^\dagger\Psi$ and the standard connection transformation law, leaving $\mathcal{D}_\mu\Psi$ covariant.

6.2 Effective action for the slow sector

Under the multiscale separation (single-band or near-degenerate subspace) and smoothness of the polarization bundle, the coarse-grained slow dynamics admits a gauge-invariant effective action built from Ψ and the curvature $\mathcal{F}_{\mu\nu}$:

$$S_{\text{eff}}[\Psi,\mathcal{A}] \;=\; \int \mathrm{d}^4x\,\Big(\Psi^\dagger\mathcal{K}(\mathcal{D})\Psi \;-\; \kappa\,\mathrm{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) \;+\; \cdots\Big), \tag{48}$$

where $\mathcal{K}(\mathcal{D})$ is an operator determined by the linearized carrier band (e.g. a Klein–Gordon-type or Dirac-type envelope operator), κ is an emergent stiffness for curvature fluctuations, and "$\cdots$" denotes higher-order and symmetry-allowed terms. In the Abelian case ($n=1$) the curvature term reduces to $F_{\mu\nu}F^{\mu\nu}$ with $F_{\mu\nu}=f_{\mu\nu}$. For comparison with electromagnetic conventions one may introduce a dimensionful potential $A_\mu^{\text{eff}} := \eta\,a_\mu$ (and analogously in the non-Abelian case), so that $F^{\text{eff}}_{\mu\nu} = \partial_\mu A^{\text{eff}}_\nu - \partial_\nu A^{\text{eff}}_\mu = \eta\,f_{\mu\nu}$. The conversion factor η fixes units and would have to be determined by matching to experiment; it is not introduced as a fundamental coupling in the microscopic substrate equations.

6.3 Triplet envelope sector and $U(3)\simeq U(1)\times SU(3)$ decomposition

If the retained band-isolated sector is a three-dimensional near-degenerate subspace (as suggested by an axis-aligned triplet on a cubic lattice), then $n=3$ and the emergent connection is $\mathcal{A}_\mu\in\mathfrak{u}(3)$. Decomposing into trace and traceless parts,

$$\mathcal{A}_\mu = \underbrace{\frac{1}{3}\mathrm{tr}(\mathcal{A}_\mu)\,I_3}_{U(1)\text{ part}} + \underbrace{\sum_{a=1}^{8}A^a_\mu\,T^a}_{SU(3)\text{ part}}, \tag{49}$$

suggests a direct route to a "QCD-like" internal gauge structure in the slow-sector EFT: the $U(1)$ part behaves electromagnetism-like, while the traceless part carries non-Abelian self-couplings through the curvature commutator term in $\mathcal{F}_{\mu\nu}$. The pairing of an Abelian factor with a non-Abelian one, and the normalization relating them, is the familiar structure of unified gauge theory [6, 22]. Crucially, for nonzero inter-axis mixing, statements about "a Berry phase per axis" are not gauge-invariant; the invariant content is the holonomy/curvature of the transported subspace (Wilczek–Zee). In that sense, the triplet sector is the natural effective setting in which a baryon-like confined mode should first be searched numerically on the cubic substrate.

EM as the symmetric (trace) direction. The $U(1)$/trace component of $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ is the projection onto the symmetric triplet direction $(1,1,1)/\sqrt{3}$:

$$\Psi_{\rm EM} \;=\; \frac{1}{\sqrt{3}}(\psi_x + \psi_y + \psi_z)\,\frac{(1,1,1)^\top}{\sqrt{3}}. \tag{50}$$

A pure-EM (charge-carrying) excitation therefore displaces all three lateral axis components *in phase* with equal weight; the traceless complement ($SU(3)$) encodes differential phase and amplitude relationships *between* the three components. As a consequence, a single-component displacement (e.g. $\psi_x \neq 0$, $\psi_y = \psi_z = 0$) projects as $\frac{1}{3}$ into the $U(1)$ trace and $\frac{2}{3}$ into the $SU(3)$-traceless sector; it is therefore not a pure-EM excitation.

Topological structure of the two sectors. The $U(1)$ and $SU(3)$ factors carry sharply different topology [34, 36, 49]: $\pi_1(U(1)) = \mathbb{Z}$ admits line vortex defects (phase winds by 2π around the core [28, 37]); $\pi_1(SU(3)) = 0$ forbids vortices; and $\pi_3(SU(3)) = \mathbb{Z}$ admits Skyrmion/instanton-type textures [4, 45]. The two sectors therefore support topologically *different* classes of localized objects: codim-2 vortex worldsheets in the EM/$U(1)$ sector versus codim-3 texture worldlines in the color/$SU(3)$ sector. Field theory provides explicit precedents both for the coexistence of these sectors and for the subtlety of their stability: a gauged $U(1)$ vortex can carry internal non-Abelian orientational moduli (semilocal and non-Abelian vortices), and such defects can be stable for dynamical rather than purely topological reasons—e.g. semilocal strings are stable on a simply connected vacuum below a critical scalar-to-vector mass ratio [1, 5, 18, 48]. This is a structural observation about the target space of the $U(3)$ field; which of these topological objects the substrate dynamics actually stabilizes, and the identification of specific objects with physical particles, are targets of the numerical program rather than conclusions of the present paper. (Open investigations under `OPEN_PROBLEMS.md` D4 and `EXPERIMENT.md`.)

Geometric phases are universal in wave systems [9, 10, 12, 39, 51]. What is specific here is the identification of the physically relevant gauge structure with the connection induced by adiabatic transport of a band-isolated wavepacket's eigenbundle on the substrate.

6.4 Dirac-envelope interpretation and the "split"

The standard Dirac field ψ bundles particle labels and gauge coupling into a single object; in the present theory that mixture is reinterpreted as follows:

- the *particle sector* corresponds to localized solitonic modes with additional internal structure (topology/geometry and its associated labels);
- the *phase/gauge sector* corresponds to the Berry/Wilczek–Zee connection induced by adiabatic transport of the band-isolated wavepacket's retained polarization subspace (its local eigenbundle).

An effective Dirac-type envelope equation is then understood as a compact coupling of these two sectors. In particular, a "charge"-like coupling parameter is not introduced as a fundamental constant of a new field; it appears as an effective conversion factor between the normalization conventions of the envelope field and the geometric connection extracted from the substrate. That a single parameter can set the relative normalization between Abelian and non-Abelian gauge sectors is itself familiar—e.g. the Abelian kinetic-mixing parameter [19, 29] and the single charge-embedding parameter that fixes the Abelian-to-non-Abelian coupling ratio in extended gauge groups [26].

7 Localized modes and self-adjoint eigenproblems

A core goal of the substrate theory is to explain particle-like behavior (localization, discrete spectra, internal "spin/charge"-like structure) as properties of confined wave states rather than point particles. In this framework, particles are extended standing-wave configurations of the substrate.

7.1 Linearization about symmetric backgrounds and self-adjointness

Let $\mathbf{X}_0(x)$ be a time-independent background embedding representing a localized deformation region. We are especially interested in backgrounds that are *spherically symmetric* in the intrinsic brane coordinates, so that invariants depend only on $r = |x|$.

Linearize the equations of motion (26) about $\mathbf{X}_0$ by writing $\mathbf{X}(x,t) = \mathbf{X}_0(x) + \boldsymbol{\eta}(x,t)$ with small perturbation $\boldsymbol{\eta}$. Because the dynamics derives from the action (24), the linearized operator is the second variation of S and is therefore self-adjoint with respect to the ρ_m-weighted L^2 inner product on Ω; orthogonality and completeness of eigenmodes follow as standard consequences.

7.2 Angular structure: scalar and vector spherical harmonics

The substrate is the 6-neighbor cubic lattice, whose point group is O_h rather than the full $SO(3)$. The long-wavelength continuum recovers approximate isotropy at scales $\gg a$, with residual cubic anisotropy of order 7.5% in the longitudinal speed at the operating point $\alpha = 0.2$ (Sprint 1). Solitons in this theory live in the Skyrme-stabilized regime of width $\gg a$, so the emergent approximate $SO(3)$ is the right symmetry to organize their angular structure, with O_h corrections entering as small splittings of $SO(3)$ multiplets.

For a spherically symmetric background, the linearized operator commutes with the action of $SO(3)$ on the intrinsic coordinates at leading order in a/ℓ_{soliton}. Consequently, eigenmodes can be chosen to transform in irreducible $SO(3)$ representations. For a scalar field this gives the familiar separation into a radial problem and an angular problem whose eigenfunctions are the spherical harmonics $Y_{\ell m}(\theta, \varphi)$.

The lateral triplet $\xi^i \in \mathbb{R}^3$, and its complex-envelope promotion $\Psi \in \mathbb{C}^3$ from Section 5, carry an internal vector index i in addition to spatial dependence. The correct angular basis is therefore the *vector spherical harmonic* basis $\mathbf{Y}^i_{JLM}$, where J is the total angular momentum of the field, L is the orbital angular momentum, and $L \in \{J-1,\ J,\ J+1\}$:

$$\xi^i(r,\theta,\varphi) \ = \ \sum_{J,L,M} f_{JLM}(r)\, \mathbf{Y}^i_{JLM}(\theta,\varphi). \tag{51}$$

The choice of (J, L) specifies how the internal index i is glued to the spatial angular coordinates — it is the choice of how the lateral triplet's $U(3)$ gauge structure (Section 5) couples to the geometry of the localized mode. Configurations with $J = 0$ are spherically symmetric only under the *combined* rotation that rotates both spatial and internal indices together; this combined rotation is the physically meaningful notion of "spherical symmetry" for a triplet field.

At soliton scale, O_h refinements enter as small splittings of $SO(3)$ multiplets. The $L = 1$ representation maps intact to the T_{1u} irrep of O_h, so vector ansätze of total orbital $L = 1$ survive without restriction. Higher-L modes split (e.g. $L = 2 \to E_g \oplus T_{2g}$) and should be expressed in cubic harmonics whenever lattice corrections matter.

This framework is descriptive, not structural: it organizes the angular content of emergent localized modes, but it does not determine which modes are physically realized. Which (J, L)

channels support stable stationary solutions is decided by the substrate dynamics together with the geometric quartic stabilization mechanism discussed in Section 3 — whose lattice-exact coefficient is $\propto k_s\alpha/a$ (vanishing as $\alpha \to 0$), so soliton stabilization requires α significantly above zero.

Appropriate radial superpositions, together with nontrivial embedding-space polarization structure, can yield torus-like, loop-like, or shell-like energy density profiles. These should be understood as standing-wave organizations of the substrate rather than as literal point particles moving on thin trajectories.

7.3 Combined particle labels at the soliton layer

A localized 3D mode is fully classified at the soliton layer by the labels

$$(J,\, P;\;\; \mathrm{SU}(3)\text{-irrep};\;\; Q_{\mathrm{U}(1)};\;\; B_{\mathrm{winding}}), \tag{52}$$

where:

- J is the total angular momentum from the emergent approximate $SO(3)$ at scales $\gg a$. On the cubic lattice this label refines to an O_h irrep at sub-leading order in $a/\ell_{\mathrm{soliton}}$.
- P is parity, the discrete element of the emergent rotation group.
- The SU(3) irrep and U(1) charge come from the $U(3)$ decomposition of the lateral triplet (Section 5).
- B_{winding} is the topological winding number of the field configuration.

These labels are not independent: the choice of how the internal triplet index couples to the spatial angular variable via the vector spherical harmonic basis sets a Clebsch–Gordan relationship between the $SO(3)$ and U(3) representations carried by the field. The canonical baryon ansatz is the *hedgehog*

$$\xi^i(\vec{x}) \;=\; f(r)\,\hat{x}^i, \tag{53}$$

with $J = 0$, $L = 1$. Here the internal index i is locked to the spatial direction $\hat{x}^i$, so the configuration is invariant only under the combined diagonal $SO(3)$. The entire nontrivial structure of the hedgehog lives in the SU(3) traceless sector: the U(1) trace $f(r)\,(\hat{x}^1 + \hat{x}^2 + \hat{x}^3)$ averages to zero on the sphere, giving a trace-neutral far field. Topological stabilization (winding $B = 1$) is achieved by extending the hedgehog into a Skyrme-twisted configuration in which the embedding-amplitude direction X^4 also rotates radially as a function of r.

The proton-versus-neutron distinction enters as a $J = 0$ trace admixture on top of the hedgehog: a nonzero scalar $L = 0$ component of ξ^i shifts the U(1) far field to proton-like, while a pure hedgehog or Skyrme-twisted hedgehog has neutron-like far field.

7.4 Amplitude scaling and energy normalization

The linear eigenproblem fixes mode shapes and frequencies but not the overall amplitude: if $\boldsymbol{\eta}$ is an eigenmode, so is $c\,\boldsymbol{\eta}$ for any scalar c. Physical amplitudes are selected only when nonlinearities are included and the total energy is fixed. For an electron-like excitation one may impose a rest-energy normalization

$$\overline{E}[\mathbf{X}] \;=\; m_e c^2, \tag{54}$$

where $\overline{E}$ denotes the conserved substrate energy associated with (24). For baryon-like modes the same logic applies with the appropriate target rest energy of the chosen species. The important structural point is unchanged: amplitude is not determined by the linear eigenproblem alone.

7.5 Confinement as geometric self-guidance

A confining mechanism is required: in a purely linear wave system, localized packets generally disperse or radiate. A useful conceptual benchmark is the toroidal electron proposal attributed to Williamson & van der Mark, which argues that ordinary gravitational curvature is far too weak to confine a Compton-scale photon-like mode and that confinement must arise from a self-consistent, excitation-specific mechanism. We adopt the *confinement requirement* while not committing to a literal double-loop trajectory (see e.g. the review discussion in [32]).

In the present substrate theory, the first candidate confinement mechanism is the brane's *geometric* self-guidance. Large-amplitude deformations modify the local Euclidean link lengths on the lattice and the induced metric in the continuum; this changes the local stress response and therefore the propagation characteristics of fluctuations around the deformed state. The crucial point is that this mechanism is already present even when the microscopic link law is Hooke-linear in the scalar extension. Constitutive refinements may strengthen or weaken localization, but the underlying nonlinear feedback comes from the same geometry that couples transverse amplitude to lateral strain.

7.6 Kinematic confinement of color

Before asking whether the substrate supports a bound colored mode, it is worth recording a structural fact about the *unbound* spectrum: the linear theory carries no colored asymptotic free state. Recall from Section 5 that the $U(1)$ trace of the lateral triplet is read as electromagnetism and the $SU(3)$ traceless content as color, both carried by the per-wavepacket complex envelope $\Psi \in \mathbb{C}^3$.

On the 6-neighbor stencil the dynamical matrix $D(k)$ is diagonal in the axis-aligned basis, with three lateral branch frequencies $h_a(k)$ whose spread is measured by the directional anisotropy $g(\hat{k}) = \sqrt{\sum_a (h_a - \bar{h})^2}/\bar{h}$. Two requirements on a freely propagating triplet wavepacket are then mutually exclusive:

- **Coherence.** A wavepacket whose three components stay phase-aligned as it propagates requires the three branches to be degenerate, $g(\hat{k}) = 0$. On the cubic lattice this holds *only* along the body-diagonal $k \parallel [111]$.
- **Color-activity.** Operationally meaningful $SU(3)$-traceless content — a finite fibre-internal holonomy distinguishing the three axis components — requires the branches to be *non-degenerate*, $g(\hat{k}) \neq 0$, i.e. propagation *off* [111]. Exactly on [111] the degenerate triplet has no preferred internal basis and the color holonomy is undefined.

No linear propagation direction satisfies both at once: off [111] the unequal branch speeds dephase the triplet over long range, while on [111] there is no resolvable color. Consequently *color charge cannot be carried to infinity by a free linear excitation.* It has support only in a nonlinear, phase-locked bound state, where the three components are held together by the geometric coupling of Section 3 rather than by spectral degeneracy. We refer to this as the *kinematic confinement of color.*

This is a statement about asymptotic states — there is no colored free particle — and not a dynamical derivation of a linear inter-quark potential, string tension, or area law, which remain outside the present scope (Section 2). It is consistent with the vanishing of the Brillouin-zone Berry/Wilczek–Zee curvature noted in Section 5: color is not a curvature on the Brillouin zone but a fibre-internal object on (x, t) that only closes around a localized worldtube. The picture carries a falsifiable, normalization-independent handle: the off-[111] fibre-holonomy ratio scales with prestress as $R(\alpha_2)/R(\alpha_1) = \big[(3 - 2\alpha_2)/\alpha_2\big]/\big[(3 - 2\alpha_1)/\alpha_1\big]$, equal to 0.3077 for $(\alpha_1, \alpha_2) = (0.2, 0.5)$; a deviation above $\sim 10\%$ would falsify the spectral-susceptibility factorization underlying it.

7.7 Why the baryon sector is the next numerical target

The cubic substrate naturally singles out three axis-associated channels. That makes the proton/neutron sector a particularly informative next target: a confined three-component standing-wave mode tests the very structure that the lattice introduces most directly. In contrast, an isolated electron-like ansatz can exist without decisively probing the triplet mixing sector.

The ansatz labels (hedgehog (53), Skyrme-twisted extension, trace admixture for proton/neutron distinction) are defined in Section 7.3. The concrete four-candidate ansatz menu used in the simulation search is recorded in `paper/baryon_simulation_roadmap.md` (Phase 2). This is not yet a derivation of QCD or of nucleon phenomenology. It is a concrete question for simulation: does the substrate support stable, localized, color-confined triplet modes at all? If the answer is no, an important bridge in the theory fails early and clearly. If the answer is yes, the cubic triplet sector becomes the natural arena for sharpening the particle interpretation.

7.8 Connection to Berry/Wilczek–Zee holonomy and spin

Localized modes can carry internal polarization structure (multi-component standing waves). The Berry/Wilczek–Zee connections of Section 5 can then be evaluated on loops surrounding the localized region or on adiabatic deformations of a localized mode family. In this way, long-range "charge"-like labels are interpreted as far-field holonomy/phase structure, while the spin-$\frac{1}{2}$ 4π aspect is attributed to spinorial holonomy of a two-level polarization subspace.

8 Bell's theorem and the retrocausal worldtube interpretation

The theory developed in Sections 3 to 7 is manifestly local and deterministic: a classical brane action whose dynamics is fixed by the embedded configuration, with all emergent objects (effective metric, gauge connections, soliton labels) derived as read-outs of the substrate state. Bell's theorem then forces an interpretive stance. The 4D-in-4D ontology of Section 3.1 commits the present program to a specific resolution, which we make explicit in this section.

8.1 The Bell constraint

Bell's theorem rules out theories that are simultaneously (i) *local* in the sense that no influence propagates outside the future lightcone of an event, (ii) *deterministic* in the sense that hidden variables fully fix outcomes given the measurement context, and (iii) *measurement-independent* in the sense that the joint distribution of detector settings and hidden state factorises over distant detectors [8, 13]. The experimental violation of Bell inequalities, by now closing the standard loopholes [24, 27, 44], then implies that at least one of (i)–(iii) must be given up by any classical-substrate completion that aims to reproduce the empirical correlations of entangled systems.

A literal superdeterministic completion — giving up (iii) by correlating detector settings with the hidden state at the Big Bang — is logically consistent but is generally regarded as explanatorily empty: it postpones rather than resolves the question of where correlations come from [31].

8.2 The retrocausal loophole and where this theory lives

The alternative consistent with the substrate ontology of Section 3.1 is the *retrocausal* loophole, which relaxes (iii) not by tying detector settings to a cosmic initial condition but by allowing the future measurement context to propagate back along the particle's worldtube to its origin in the shared past of the entangled pair. Spacelike correlations are then not action-at-a-distance but *continuity of a single extended 4D object* whose world-volume branches at a V-vertex in the past. The present theory places itself in the same interpretive family as the two-state-vector formalism of Aharonov and collaborators [2, 3], Cramer's transactional interpretation [14], the retrocausal models of Price and Wharton [40], and Sutherland's time-symmetric Bohmian model [47].

This stance fits the 4D-in-4D ontology naturally. The brane action is time-symmetric: its Lorentzian sign structure (Section 3.1) picks out a *kind* of direction (timelike vs spacelike) but not a *direction* along the timelike axis (past-to-future vs future-to-past). The arrow of time is therefore not a property of the substrate; it is selected at the soliton level by the chirality of soliton solutions.

8.3 Causality from soliton chirality; matter and antimatter

In this stance, the empirical arrow of time is a property of *solitons*, not of the substrate. Matter solitons are forward-propagating worldtubes; antimatter solitons are backward-propagating worldtubes. This realises the Feynman–Stueckelberg interpretation of antiparticles as time-reversed particles [20, 46] at the level of substrate soliton solutions rather than as an interpretive remark on a Lorentz-invariant field theory. The cosmological baryon–antibaryon asymmetry is, on this reading, a *geometric initial-condition asymmetry*: matter and antimatter solitons emerge from the Big Bang vertex propagating in opposite time directions along the brane's selected timelike axis, and our universe contains only the forward-propagating branch. This differs structurally from Sakharov-condition baryogenesis [43] in which the same time direction hosts both matter and antimatter, with a small dynamical asymmetry favouring matter.

8.4 Entanglement as V-branching worldtubes

An entangled pair of solitons, on the present reading, is *one* branching 4D world-volume whose worldtube splits at a V-vertex in the shared past. Each branch propagates forward to a measurement event; the boundary condition imposed by the measurement on one branch is communicated to the other branch via continuity through the shared V-vertex. The result is statistical correlation that is local along each worldline (no superluminal influence on any spacelike slice the lab observer can resolve) and deterministic given the full 4D configuration, while still violating Bell inequalities because the joint measurement context affects the past of both branches via the V-vertex. The 4D world-volume picture promotes "spooky action at a distance" to "ordinary continuity of one extended object".

9 Discussion

9.1 What is fixed by the present theory

At the level stated here, the theory fixes five structural commitments: (i) a 4D cubic brane lattice embedded with codimension zero in a symmetric 4D Euclidean ambient, with the timelike direction selected by the Lorentzian sign structure of the brane action (Section 3.1), (ii) a geometric amplitude–lateral coupling generated by Euclidean link lengths / the induced metric, applied symmetrically across the four ambient directions, (iii) per-wavepacket band isolation whose adiabatically transported eigenbundle carries Berry/Wilczek–Zee connection data, (iv) the interpretation of particles as localized standing-wave modes rather than point objects, and (v) a retrocausal worldtube reading (Section 8) in which causality is selected by soliton chirality and entanglement is the continuity of a single 4D world-volume that branches at a V-vertex in the past. These commitments do not yet close the theory, but they sharply constrain what kinds of completions are still allowed.

9.2 Why the mathematical bridges must remain explicit

The present framework only becomes interesting if its bridges are written explicitly enough to be attacked in actual calculation. For that reason, the continuum action, the lattice substrate, the contraction channel, the per-wavepacket band-isolation framing, the holonomy interpretation, and the localized-mode program should not be treated as optional stylistic embellishments. They are the working architecture of the theory. Removing such bridges would make the paper shorter, but it would also make the theory less testable and less coherent.

9.3 Microscopic cubicity and macroscopic isotropy

The dual-observer mechanism by which operational isotropy can emerge despite a cubic UV substrate is stated canonically in Section 3.2. The discussion-level point is that this is not guaranteed. If different branches, wavelengths, or polarization sectors feel the lattice differently, residual anisotropy and birefringence survive as observable signatures. The theory therefore lives or dies on a quantitative statement: whether one can identify a dominant observer sector in which anisotropy is sufficiently suppressed and sufficiently universal, with residual birefringence as the failure signature.

9.4 Axis-aligned triplet phases and non-Abelian transport

The axis-aligned triplet idea is best understood as a basis adapted to the cubic substrate. On that substrate it is natural to organize a band-isolated wavepacket triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top$ aligned with the lattice axes, with each component arising from per-wavepacket complex-envelope demodulation of the corresponding lateral channel. If inter-axis mixing is negligible, each component carries an approximately independent rank-1 Berry phase. For any nonzero mixing, however, the physical eigenmodes are mixtures of axis labels and the invariant content is the Wilczek–Zee holonomy of the transported subspace, not a separate gauge-invariant phase "of an axis." This distinction (independent-$U(1)^3$ vs Wilczek–Zee $U(3)$) is what makes the triplet sector non-trivially different from three decoupled EM channels, and it is the theoretical reason the baryon sector is the decisive numerical target (see Section 7.7).

9.5 Contact with lattice gauge ideas

Standard lattice QCD uses the lattice primarily as a regulator for a continuum gauge theory. Here the interpretive direction is reversed: the lattice is ontic, while gauge structure is read out as connection data of band-isolated wavepacket eigenbundles. Despite this difference, much of the diagnostic machinery transfers directly. Berry connections and curvatures can be computed on a discretized Brillouin zone using link variables and plaquettes, allowing gauge-invariant comparison of Abelian and non-Abelian transport sectors. The analogy is therefore methodological rather than ontological.

9.6 Related emergent-$U(1)$ constructions on 3D cubic substrates

It is worth situating the present construction against peer-reviewed precedents in which a 3D cubic-lattice substrate is known to host an emergent $U(1)$ gauge sector at low energies.

Han and Kivelson [25] construct a 3D cubic lattice with bond-centered atomic displacements coupled to electrons and prove that strong electron–phonon coupling drives the system into an ice-rule constrained "phononic resonating-valence-bond" phase whose effective theory near the Rokhsar–Kivelson point is a $U(1)$ gauge theory of the form $\mathcal{H} = \frac{1}{2\tilde{\varepsilon}}|\boldsymbol{d}|^2 + \frac{1}{2\tilde{\mu}}(\kappa^2|\nabla\cdot\boldsymbol{d}|^2 + |\nabla\times\boldsymbol{a}|^2)$, with gapless transverse-optical phonons playing the role of the emergent photon and deconfined irrationally charged excitations playing the role of matter. In substrate class their model is the closest published analogue of the present brane lattice: both live on a 3D cubic geometry of coupled oscillators and both produce a gapless $U(1)$ sector at low energies. The mechanisms, however, are distinct. There, gauge invariance is enforced as a low-energy consequence of an ice-rule constraint on bond-centered displacements generated by an electronic sector; here, the substrate is purely mechanical and gauge structure appears as the Berry/Wilczek–Zee holonomy of an isolated polarization band along the carrier eigenbundle. The shared conclusion is the qualitative plausibility of an emergent gapless $U(1)$ in 3D cubic oscillator networks; the divergence is that the present theory extends the same logic to a near-degenerate triplet whose non-Abelian Wilczek–Zee transport realizes $U(1) \times SU(3)$ rather than $U(1)$ alone (cf. Section 5.6), with the prestretch parameter α running the crossover.

A second precedent is the $U(1)$ quantum spin liquid expected in 3D pyrochlore spin ice [23]. There a local 2-in/2-out ice-rule constraint at each tetrahedral vertex produces an emergent compact $U(1)$ gauge theory with a gapless "photon" mode and deconfined magnetic monopole excitations. The mechanism is again constraint-driven and operates on spin rather than on lattice displacement, but it reinforces the same structural lesson: 3D substrates with a vertex-local conservation law are a natural setting for emergent electromagnetism.

Non-Abelian gauge structure, by contrast, has been realized in *classical wave* media outside the condensed-matter setting. Chen et al. [11] obtain a synthetic $SU(2)$ gauge field for optical waves directly from the anisotropy of a homogeneous medium, and Kumar et al. [35] design continuous elastic waveguides whose holonomy—acquired by *crossing* band degeneracies rather than by maintaining them as in the Wilczek–Zee construction used here—transfers excitation between rods in a concatenation-order-dependent (hence non-commuting) way. These establish that non-Abelian holonomy on classical elastic/optical waves is physically realizable, which lends qualitative support to the triplet-sector claim of Section 5.6. The distinction is one of provenance: in both works the gauge response is *engineered* into the medium by a spatially designed anisotropy or coupling profile and demonstrated as a mode-transport effect, whereas here the non-Abelian sector is meant to arise spontaneously from a single ontic substrate that must *also* host the $U(1)$ sector, the soliton matter sector, and the gravity-like channel together.

What none of these constructions provide is (i) non-Abelian internal structure arising spontaneously from the same unengineered substrate that hosts the matter and gravity sectors, (ii) a Lorentz-respecting wide-soliton sector decoupled from a Peierls–Nabarro lattice scale, or (iii) a gravity-like channel sourced by transverse excitation through the induced metric. These are precisely the points where the present theory takes on commitments that are absent in Han and Kivelson [25], Gingras and McClarty [23], Chen et al. [11], and Kumar et al. [35], and that must therefore be tested on their own merits.

9.7 Gravity as contraction from transverse excitation

The full mechanism — Pythagorean spring-length effect, $g_{ij} = h_\star^2\, \delta_{ij} + \partial_i u\, \partial_j u$, and the resulting inward lateral pull under homogeneous prestretch — is derived in Section 3.9.

What remains open is the quantitative reduction. One would like to derive a Newtonian limit in which an effective scalar potential built from coarse-grained strain or energy density obeys a Poisson-type equation with a universal coupling constant, and then relate that constant to G. On the simulation side, the central check is straightforward: create a localized transverse excitation, measure the induced steady contraction field, and quantify its action on long-wavelength probe packets.

9.8 Quantum phenomenology without fundamental probability

By construction, the underlying dynamics is deterministic. The probabilistic character of quantum measurements is attributed to the interaction of localized modes with wave packets that are constrained by Fourier localization and by the finite Brillouin zone, together with nonlinear exchange events in which small phase and amplitude differences can decide whether energy transfer occurs. In this view, apparent point-like interactions reflect the structure of wave packets and nonlinear exchange, not the existence of fundamental point particles.

9.9 Priority open problems

Several problems remain before the theoretical framework could be regarded as a mature replacement attempt:

- establishing a universal observer sector with sufficiently suppressed long-wavelength anisotropy,
- deriving a quantitative contraction-channel reduction and matching it to an effective gravitational constant,
- identifying parameter regimes that simultaneously support band-isolated adiabatic wavepackets and robust localized modes,
- determining whether the cubic triplet sector really supports stable baryon-like confined modes,
- clarifying how far-field Abelian holonomy and internal non-Abelian transport map onto observed particle labels,
- understanding multi-particle states and interaction processes in the deterministic substrate.

The paper is now best read as a sharpened core theory plus a numerical research program whose first decisive test is the existence or failure of stable triplet-confined baryon-like modes.

10 Conclusion

We presented a deterministic substrate theory in which observable physics is treated as arising from waves on a 4D brane lattice embedded with codimension zero in a 4D Euclidean ambient. The ambient is fully symmetric; the timelike direction is selected by the Lorentzian sign structure of the brane action, and the gauge/gravity split is the inside observer's view after picking that direction. The continuum model is stated as the long-wavelength effective description of the cubic substrate on a spacelike slice, using an isotropic hyperelastic action built from the induced metric and a homogeneous prestretch parameter α as the analytic baseline. The bare cubic stencil must still pass the separate dispersion/isotropy tests, because the Cauchy relation alone does not guarantee an isotropic leading-order acoustic tensor.

A central clarification of the present revision is that the essential nonlinearity is already geometric. Even when the microscopic link energy is Hooke-linear in scalar extension, the full Euclidean link length in $\mathbb{R}^4$ and the induced metric generate nonlinear amplitude–lateral coupling. That same mechanism is intended to underwrite contraction, self-guidance, and localization.

Applying the complex-envelope description per band-isolated wavepacket, we constructed Berry and Wilczek–Zee connections from adiabatic eigenbundle transport and interpreted them as the natural gauge variables of the per-wavepacket coarse-grained envelope description. On a cubic lattice, an axis-aligned triplet sector provides the natural candidate internal arena in which a non-Abelian transport structure can emerge. Localized particle-like states were then posed as self-adjoint eigenproblems around symmetric backgrounds, with confinement attributed to geometric self-guidance rather than to fundamental point charges or externally imposed potentials.

The most important consequence is methodological. The theory is now stated in a way that makes its next decisive test clear: the search for stable, localized, three-component baryon-like modes on the cubic substrate. Such a calculation would simultaneously probe the geometric confinement mechanism, the axis-aligned triplet structure, and the viability of coarse-grained isotropy. Whether that program succeeds or fails is more important than further verbal extrapolation. The immediate task is therefore to turn the consolidated theory into direct computation.

The interpretive commitment that closes the present paper — the retrocausal worldtube reading of Section 8 — is not a free embellishment. It is forced on any local + deterministic substrate by Bell's theorem, and it shapes how the simulation programme should ultimately read its own results: matter and antimatter as forward- and backward-propagating soliton worldtubes, and entanglement as continuity of one 4D object that branches at a V-vertex in the past.

11 Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x,t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	Slicing parameter along the inside observer's timelike direction ($t := x^4$); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g^0_{ij}	Isotropic reference metric: $g^0_{ij} = \ell_0^2\, \delta_{ij}$ (stress-free scale ℓ_0).
$h_\star$	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0 / h_\star \in (0, 1]$.
$\widehat{g}$	Mixed relative metric: $\widehat{g} = (g^0)^{-1} g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$.
λ, μ	Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1-\alpha) k_s / a$, $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
$P^{\ i}_{A}$	First Piola stress: $P^{\ i}_{A} = \partial W / \partial(\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_\star$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
$\boldsymbol{\epsilon}$	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = \mathrm{i}\langle u \,\vert\, \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n-dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - \mathrm{i}[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ: $\gamma_\Gamma = \oint_\Gamma a_\mu \, \mathrm{d}x^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\mathrm{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.
$R^l_p \in \mathbb{R}^4$	4D node position in the explicit lattice: $p = (i, j, k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.
k_s	Central-force spring stiffness of the 6-neighbor spacelike stencil.
m	Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$).
Δt	Temporal lattice step (timelike link length in the explicit 4D lattice).
$\mathcal{N}_s$	6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$.
$\mathcal{N}_t$	Temporal neighbor stencil: $\{l \pm 1\}$.
c_L	Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2 / m$.
c_T	Long-wavelength transverse wave speed: $c_T^2 = (1-\alpha)\, k_s a^2 / m$.

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